MATH MA'AM ACADEMY

Where Every Subject Clicks.

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UNIT 3: TYPES OF FORCE

Expanded revision notes based on the supplied textbook pages, with additional concepts, examples, formulae and application facts for Homi Bhabha preparation.

1. WHAT IS FORCE?

Force is a push or pull acting on an object due to its interaction with another object. Force is a vector quantity: it has both magnitude and direction.

- A force can start or stop motion, change the speed of a moving object, change its direction, or change its shape and size.
- A force may act only when objects touch, or it may act from a distance.
- SI unit of force: newton (N). CGS unit: dyne. 1 N = 100,000 dyne = 10^5 dyne.
- One newton is the force that produces an acceleration of 1 m/s² in a mass of 1 kg.

Effects of force

- Change the state of rest: pushing a stationary trolley makes it move.
- Change speed: applying brakes slows a moving bicycle; pedalling harder can increase its speed.
- Change direction: a bat changes the direction of a moving ball.
- Change shape: squeezing a sponge or stretching a spring changes its shape.

2. CLASSIFICATION OF FORCES

Contact forces	Non-contact forces
Act only when interacting objects are in physical contact.	Act without direct physical contact between interacting objects.
Examples: muscular force and frictional force.	Examples: gravitational, magnetic and electrostatic forces.

3. CONTACT FORCES

A. Muscular force

The force applied by muscles of humans and animals is called muscular force. It is a contact force because the body or part of the body must touch the object to apply the force.

- Examples: lifting a schoolbag, pushing a cart, kicking a football, pulling a rope and an animal pulling a plough.
- Muscular force can produce movement or change the shape of an object.

B. Frictional force

Friction is a force that opposes relative motion, or the tendency of relative motion, between surfaces in contact. It acts along the surfaces and opposite to the direction of relative sliding.

- Friction acts between solid surfaces, and also between objects moving through fluids such as air and water (fluid resistance or drag).
- Friction can slow down or stop moving objects. It also produces heat.
- Friction provides grip: we can walk without slipping, write with a pencil, hold objects and apply brakes because of friction.
- Friction can cause wear and tear in machine parts and wastes some energy as heat.

Factors affecting friction

- Nature and roughness of the surfaces: rougher surfaces often produce greater friction than smoother surfaces.
- How strongly the surfaces are pressed together: greater normal force generally increases dry friction.
- For many ordinary situations, friction is not determined simply by the apparent area of contact.

Increasing and reducing friction

- Increase friction: treads on tyres, grooves on shoe soles, rough handles, spreading sand on slippery roads.
- Reduce friction: lubricants such as oil or grease, polishing surfaces, using wheels or rollers, and ball bearings.
- Air resistance is reduced in streamlined vehicles such as aeroplanes and racing cars.

4. NON-CONTACT FORCES

A. Gravitational force

Gravitational force is the attractive force between any two objects that have mass. It acts over a distance and is always attractive.

- Every object with mass attracts every other object with mass. The force becomes stronger when masses are larger and weaker when the distance between their centres increases.
- Earth's gravitational force pulls objects towards its centre. This force gives objects their weight and keeps the Moon in orbit around Earth and planets in orbit around the Sun.
- Gravity acts even when objects are not touching one another.

B. Magnetic force

Magnetic force is the force exerted by a magnet on another magnet or on magnetic materials such as iron, nickel and cobalt.

- Like magnetic poles repel: north–north and south–south repel.
- Unlike magnetic poles attract: north–south attract.
- A magnet can attract suitable magnetic materials without touching them; therefore magnetic force is a non-contact force.
- Not all metals are strongly attracted to ordinary magnets. Iron, nickel and cobalt are common examples of ferromagnetic materials.

C. Electrostatic force

Electrostatic force is the force between electrically charged objects. It can act from a distance.

- Like charges repel and unlike charges attract.
- A plastic comb rubbed through dry hair can become electrically charged and attract small pieces of paper.
- Rubbing can transfer electrons from one material to another, leaving the materials electrically charged.
- Electrostatic precipitators in industries use electric charges to remove fine particles from smoke and reduce air pollution.

5. MASS AND WEIGHT

Mass	Weight
Amount of matter in an object; a measure of its inertia.	Gravitational force acting on an object.
Scalar quantity.	Vector quantity; directed towards the attracting body (near Earth, downward).
SI unit: kilogram (kg).	SI unit: newton (N).
Does not change when an object is taken from Earth to the Moon (ignoring tiny relativistic effects).	Changes when gravitational field strength changes.
Commonly measured using a beam balance or electronic balance.	Can be measured using a spring balance.

Formula for weight

Weight (W) = Mass (m) × gravitational field strength (g)

W = mg

- Near Earth's surface, g is approximately 9.8 N/kg (often rounded to 10 N/kg for school calculations).
- On the Moon, gravitational field strength is approximately one-sixth of Earth's. An object weighs about one-sixth as much on the Moon as on Earth, but its mass remains the same.
- Example: an object of mass 6 kg has weight about 60 N on Earth if g = 10 N/kg. Its weight on the Moon is about 10 N.
- A spring balance measures force/weight, not mass directly. A beam balance compares masses.
- Gravity varies slightly from place to place. It is generally greater at the poles than at the equator; therefore an object's weight is slightly greater at the poles.

6. PRESSURE

Pressure is the force acting normally (perpendicularly) per unit area of a surface.

Pressure = Force ÷ Area or P = F/A

- SI unit: pascal (Pa). 1 Pa = 1 N/m^2.
- For the same force, smaller area produces greater pressure; larger area produces smaller pressure.
- For the same area, greater force produces greater pressure.

Everyday applications of pressure

- Knives, blades, needles and pins have sharp edges or tips to produce high pressure over a small area.
- School bags have broad straps to spread the force over a larger area and reduce pressure on the shoulders.
- Camels have broad feet that spread their weight over a larger area, helping them walk on sand.
- A pointed nail enters wood more easily than a blunt nail because its tip produces greater pressure.

7. ATMOSPHERIC PRESSURE

The air surrounding Earth has mass and is pulled by gravity. The weight of the air produces atmospheric pressure on Earth's surface and on objects.

- Atmospheric pressure acts in all directions, not only downwards.
- At sea level, standard atmospheric pressure is about 101,325 Pa, approximately 1.01×10^5 Pa or 1 atmosphere.
- Atmospheric pressure is measured using a barometer.
- Atmospheric pressure generally decreases as altitude increases because there is less air above a location.
- A drinking straw works because reducing pressure inside the straw allows atmospheric pressure on the liquid surface to push liquid up the straw.

8. PRESSURE EXERTED BY LIQUIDS

- Liquids exert pressure on the bottom and walls of their container.
- Liquid pressure increases with depth. At greater depth, a larger column of liquid is above the point.
- At the same depth in the same liquid at rest, pressure is the same in all directions.
- At a given depth, liquid pressure depends on the liquid's density and gravitational field strength. It does not depend on the shape of the container.
- Deep-sea divers need special equipment because water pressure becomes very large at great depths.
- Dams are made thicker near the bottom because water pressure is greater at greater depths.

9. IMPORTANT COMPARISONS

Concept	Key distinction
Contact vs non-contact	Contact requires physical touch; non-contact acts across a distance.
Mass vs weight	Mass is amount of matter; weight is gravitational force.
Force vs pressure	Force is a push or pull; pressure is force per unit area.
Like vs unlike magnetic poles	Like poles repel; unlike poles attract.
Like vs unlike electric charges	Like charges repel; unlike charges attract.
Liquid pressure vs depth	Pressure increases as depth increases.
Friction vs motion	Friction opposes relative motion or its tendency between contacting surfaces.

10. QUICK FORMULA AND FACT SHEET

Quantity / fact	Formula or value
Force	SI unit = newton (N); CGS unit = dyne
Conversion	1 N = 10^5 dyne
Weight	W = mg
Approximate Earth g	9.8 N/kg (often use 10 N/kg)
Approximate Moon gravity	About 1/6 of Earth's
Pressure	P = F/A
Pressure unit	1 Pa = 1 N/m^2
Standard atmospheric pressure	About 1.01 × 10^5 Pa at sea level

11. APPLICATION-BASED FACTS TO REMEMBER

- A football slows down on grass mainly because of friction and air resistance.
- A magnet attracts an iron nail without touching it: magnetic, non-contact force.
- A charged comb attracts paper bits: electrostatic, non-contact force.
- An apple falls towards Earth due to gravitational force.
- A person pushes a wall but it does not move: force is applied, but there is no resulting displacement of the wall.
- A sharp knife cuts more easily because the same applied force acts over a smaller area, creating greater pressure.
- A diver experiences greater water pressure at greater depth.
- A spring balance reading changes on the Moon because weight changes; the object's mass does not.
- Friction is not always harmful: without it, walking, gripping and braking would be difficult.

For Homi Bhabha preparation, practise identifying the type of force in everyday situations and applying the relationships W = mg and P = F/A.